

8.5 Partial Fractions

We can add fractions as follows.

$$\frac{5}{x+3} + \frac{2}{x-1} = \frac{5(x-1) + 2(x+3)}{(x+3)(x-1)}$$

$$\frac{5}{x+3} + \frac{2}{x-1} = \frac{7x+1}{x^2+2x-3}$$



Partial Fractions is about going in the opposite direction.

Suppose we need to evaluate $\int \frac{7x+1}{x^2+2x-3} dx$.

It becomes much simpler once we notice that it is equivalent to :

$$\int \left(\frac{5}{x+3} + \frac{2}{x-1} \right) dx = 5 \ln|x+3| + 2 \ln|x-1| + C$$

Let's see how to expand $\frac{7x+1}{x^2+2x-3}$ into partial fractions.

First we factor the denominator :

$$x^2 + 2x - 3 = (x+3)(x-1)$$

The partial fractions decomposition will have the form :

$$\frac{7x+1}{(x+3)(x-1)} = \frac{A}{x+3} + \frac{B}{x-1}$$

where A and B are to be determined.

There are various methods that can be used to find A and B. We will show two methods. Both methods start by observing that :

$$7x+1 = A(x-1) + B(x+3)$$

Method ① Plug values for x .

$$7x + 1 = A(x - 1) + B(x + 3)$$

$$\text{Set } x = 1 \rightarrow 7(1) + 1 = 0 + B(1 + 3)$$

$$\rightarrow 8 = 4B$$

$$\rightarrow \boxed{B = 2}$$

$$\text{Set } x = -3 \rightarrow 7(-3) + 1 = A(-4) + 0$$

$$\rightarrow -20 = -4A$$

$$\rightarrow \boxed{A = 5}$$

Method ② Expand and match coefficients

$$7x + 1 = A(x - 1) + B(x + 3)$$

$$7x + 1 = Ax - A + Bx + 3B$$

$$\underline{7x + 1} = \underline{(A + B)x} + \underline{(-A + 3B)}$$

$$\Rightarrow \begin{cases} A + B = 7 \\ -A + 3B = 1 \end{cases} \xrightarrow{\text{solve}} \begin{cases} A = 5 \\ B = 2 \end{cases}$$

We can expand a quotient of Polynomials

$$\frac{N(x)}{D(x)} \text{ into partial fractions as}$$

long as : (Numerator degree) < (Denominator degree)

If (Numerator degree) $\geq$ (Denominator degree)

We can use a long division to get:

$$\frac{N(x)}{D(x)} = (\text{A Polynomial}) + \frac{N_1(x)}{D(x)}$$

where : [degree of $N_1(x)$] < [degree of $D(x)$]

We can then use partial fractions to

expand : $\frac{N_1(x)}{D(x)}$.

ex: ① $\int \frac{2x^3 - 3x^2 - 3x - 10}{x^2 - 2x - 3} dx$

Step ① Long Division

We need a Long Division since: $\left(\begin{smallmatrix} \text{Num.} \\ \text{degree} \end{smallmatrix} \right) \geq \left(\begin{smallmatrix} \text{Denom.} \\ \text{degree} \end{smallmatrix} \right)$

$$\begin{array}{r} x^2 - 2x - 3 \overline{) 2x^3 - 3x^2 - 3x - 10} \\ \underline{2x^3 - 4x^2 - 6x} \\ x^2 + 3x - 10 \\ \underline{x^2 - 2x - 3} \\ 5x - 7 \end{array}$$

$$\Rightarrow \int \left[2x + 1 + \frac{5x - 7}{x^2 - 2x - 3} \right] dx$$

requires P.F.

Step ② Expand $\frac{5x - 7}{x^2 - 2x - 3}$ into P.F.

$$\text{Factor : } x^2 - 2x - 3 = (x+1)(x-3)$$

$$\frac{5x-7}{(x+1)(x-3)} = \frac{A}{x+1} + \frac{B}{x-3}$$

$$\Rightarrow 5x-7 = A(x-3) + B(x+1)$$

Now, we plug values for x to find A, B .

Note : Pick $x=3$ to make $(x-3)=0$
Pick $x=-1$ to make $(x+1)=0$

$$x=3 \rightarrow 15-7 = 0+4B \rightarrow B=2$$

$$x=-1 \rightarrow -5-7 = -4A+0 \rightarrow A=3$$

step ③ Integrate

$$\int \left(2x+1 + \frac{3}{x+1} + \frac{2}{x-3} \right) dx$$

$$= x^2 + x + 3\ln|x+1| + 2\ln|x-3| + C$$

We often encounter integrals such as:

$$\int \frac{1}{3x+4} dx,$$

You can do a substitution $u = 3x+4$
to evaluate it.

$$\begin{aligned} du &= 3 dx \\ \rightarrow \frac{1}{3} du &= dx \end{aligned}$$

$$\begin{aligned} \int \frac{1}{3x+4} dx &= \frac{1}{3} \int \frac{1}{u} du \\ &= \frac{1}{3} \ln|3x+4| + C \end{aligned}$$

To go faster, you can observe that in general:

$$\int \frac{1}{ax+b} dx = \frac{1}{a} \ln|ax+b| + C$$

for $a \neq 0$

ex: ② $\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx$

No need for Long division for this one.

Factor the denominator

$$2x^3 + 3x^2 - 2x = x(2x-1)(x+2)$$

Expand as P.F.

$$\frac{x^2 + 2x - 1}{x(2x-1)(x+2)} = \frac{A}{x} + \frac{B}{2x-1} + \frac{C}{x+2}$$

$$\Rightarrow x^2 + 2x - 1 = A(2x-1)(x+2) + Bx(x+2) + Cx(2x-1)$$

Plug values for x

$$\text{set } x=0 \rightarrow -1 = A(-1)(2) + 0 + 0 \rightarrow A = 1/2$$

$$\text{set } x=-2 \rightarrow -1 = 0 + 0 + 10C \rightarrow C = -1/10$$

$$\text{set } x=1/2 \rightarrow \frac{1}{4} = 0 + \frac{5}{4}B + 0 \rightarrow B = 1/5$$

Integrate

$$\int \left[\frac{1/2}{x} + \frac{1/5}{2x-1} + \frac{(-1/10)}{x+2} \right] dx$$

K is the Constant of integration

$$= \frac{1}{2} \ln|x| + \frac{1}{10} \ln|2x-1| - \frac{1}{10} \ln|x+2| + K$$

Repeated Linear terms

If we encounter something like :

$$\frac{2x+1}{(x+1)(x+5)^3}, \text{ its P.F. decomposition}$$

has the form :

$$\frac{2x+1}{(x+1)(x+5)^3} = \frac{A}{x+1} + \underbrace{\frac{B}{x+5} + \frac{C}{(x+5)^2} + \frac{D}{(x+5)^3}}_{\text{for } (x+5)^3}$$

In general, if the denominator includes

$(ax+b)^n$, the P.F. will include the

n terms :

$$\frac{A_1}{ax+b} + \frac{A_2}{(ax+b)^2} + \dots + \frac{A_n}{(ax+b)^n}$$

ex: (3) $\int \frac{5x^2 + 2x + 6}{x^3 + 2x^2 + x} dx$

No need for long division.

Factor: $x^3 + 2x^2 + x = x(x^2 + 2x + 1)$
 $= x(x+1)^2$

P.F.: $\frac{5x^2 + 2x + 6}{x(x+1)^2} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$

$$\Rightarrow 5x^2 + 2x + 6 = A(x+1)^2 + Bx(x+1) + Cx$$

Now, plug three values for x to find A, B, C .

Set $x=0 \rightarrow 6 = A + 0 + 0 \rightarrow A = 6$

Set $x=-1 \rightarrow 9 = 0 + 0 - C \rightarrow C = -9$

Set $x=1 \rightarrow 13 = 4A + 2B + C \rightarrow 13 = 24 + 2B - 9$
 $\rightarrow B = -1$

Integrate

$$\int \left[\frac{6}{x} - \frac{1}{x+1} - \frac{9}{(x+1)^2} \right] dx$$

$$= \boxed{6 \ln|x| - \ln|x+1| + \frac{9}{x+1} + K}$$

ex: ④ $\int \frac{2x}{(x+1)(x+2)^3} dx$

P.F. $\frac{2x}{(x+1)(x+2)^3} = \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{(x+2)^2} + \frac{D}{(x+2)^3}$

$$\Rightarrow 2x = A(x+2)^3 + B(x+1)(x+2)^2 + C(x+1)(x+2) + D(x+1)$$

set $x = -2 \rightarrow -4 = 0 + 0 + 0 - D \rightarrow \boxed{D = 4}$

set $x = -1 \rightarrow -2 = A + 0 + 0 + 0 \rightarrow \boxed{A = -2}$

There's no other "nice" values of x , Pick any two other values.

set $x = 0 \rightarrow 0 = 8A + 4B + 2C + D \rightarrow 0 = -16 + 4B + 2C + 4$
 $\rightarrow 4B + 2C = 12 \rightarrow \boxed{2B + C = 6}$

set $x = 1 \rightarrow 2 = 27A + 18B + 6C + 2D \rightarrow 2 = -54 + 18B + 6C + 8$
 $\rightarrow 18B + 6C = 48 \rightarrow \boxed{3B + C = 8}$

$$\begin{cases} 2B + C = 6 \\ 3B + C = 8 \end{cases} \xrightarrow{\text{solve}} \begin{cases} B = 2 \\ C = 2 \end{cases}$$

$$\int \left[\frac{-2}{x+1} + \frac{2}{x+2} + \frac{2}{(x+2)^2} + \frac{4}{(x+2)^3} \right] dx$$

$$= \boxed{-2 \ln|x+1| + 2 \ln|x+2| - 2(x+2)^{-1} - 2(x+2)^{-2} + K}$$

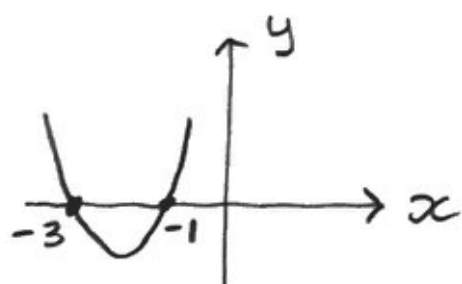
Irreducible quadratics

$x^2 - 4$ can be factored as $(x-2)(x+2)$

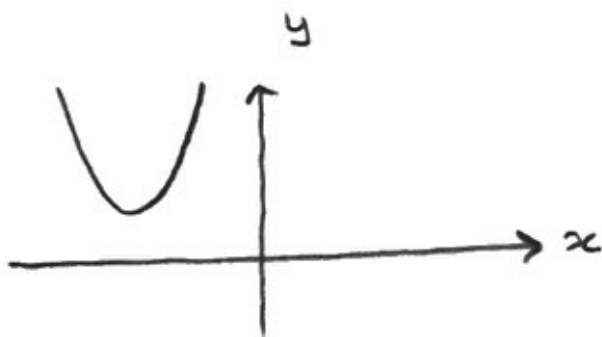
$x^2 + 4$ can't be factored, it is irreducible.

$x^2 + 4x + 3$ can be factored as $(x+3)(x+1)$

$x^2 + 4x + 13$ can't be factored.



$$y = x^2 + 4x + 3$$



$y = x^2 + 4x + 13$ has no real roots.

In general: $ax^2 + bx + c$

is irreducible if and only if

$$b^2 - 4ac < 0$$

P.F. with irreducible quadratics

To expand $\frac{2x+1}{(x+3)(x^2+4)}$ use :

$\underbrace{(x^2+4)}_{\text{irreducible}}$

$$\frac{2x+1}{(x+3)(x^2+4)} = \frac{A}{x+3} + \frac{Bx+C}{x^2+4}$$

In general, if the denominator includes a factor of the form $(ax^2+bx+c)^n$ for an irreducible quadratic: ax^2+bx+c , the P.F. decomposition must include:

$$\frac{B_1x+C_1}{ax^2+bx+c} + \frac{B_2x+C_2}{(ax^2+bx+c)^2} + \dots + \frac{B_nx+C_n}{(ax^2+bx+c)^n}.$$

Every polynomial can be factored into factors of the form : $(\underbrace{px+q}_{\text{Linear}})^m$ or $(\underbrace{ax^2+bx+c}_{\text{irreducible}})^n$.

This follows from the

Fundamental Theorem of Algebra.

ex: ⑤ $\int \frac{3x^2 + 5x + 8}{x^3 + 4x} dx$

No need for long division.

Factor : $x^3 + 4x = x \underbrace{(x^2 + 4)}_{\text{irreducible}}$

P.F. : $\frac{3x^2 + 5x + 8}{x(x^2 + 4)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4}$

$$\begin{aligned}\Rightarrow 3x^2 + 5x + 8 &= A(x^2 + 4) + (Bx + C)x \\ &= Ax^2 + 4A + Bx^2 + Cx \\ &= (A+B)x^2 + Cx + 4A\end{aligned}$$

Match coefficients

$$\begin{cases} A+B = 3 \rightarrow B=1 \\ C = 5 \\ 4A = 8 \rightarrow A=2 \end{cases}$$

To integrate :

$$\frac{Bx + C}{x^2 + 4} \text{ split it}$$

into two integrals.

$$\int \left(\frac{2}{x} + \frac{x}{x^2 + 4} + \frac{5}{x^2 + 4} \right) dx$$

$$= \boxed{2 \ln|x| + \frac{1}{2} \ln(x^2 + 4) + \frac{5}{2} \operatorname{Arctan}\left(\frac{x}{2}\right) + K}$$

ex: ⑥ $\int \frac{2x+1}{x^4+3x^2-4} dx$

No need for long division.

Factor completely the denominator

$$\begin{aligned} x^4+3x^2-4 &= (x^2-1)(x^2+4) \\ &= (x-1)(x+1) \underbrace{(x^2+4)}_{\text{irreducible}} \end{aligned}$$

P.F.

$$\frac{2x+1}{(x-1)(x+1)(x^2+4)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{Cx+D}{x^2+4}$$

$$\begin{aligned} \Rightarrow 2x+1 &= A(x+1)(x^2+4) + B(x-1)(x^2+4) + \dots \\ &\quad \dots + (Cx+D)(x-1)(x+1) \end{aligned}$$

$$\text{set } x = 1 \rightarrow 3 = 10A + 0 + 0 \rightarrow A = 3/10$$

$$\text{set } x = -1 \rightarrow -1 = 0 - 10B + 0 \rightarrow B = 1/10$$

$$\text{set } x = 0 \rightarrow 1 = 4A - 4B - D$$

$$\rightarrow D = 4A - 4B - 1 = \frac{12}{10} - \frac{4}{10} - 1 = -\frac{1}{5}$$

$$\text{set } x = 2 \rightarrow 5 = 24A + 8B + 6C + 3D$$

$$\rightarrow 5 = 24\left(\frac{3}{10}\right) + 8\left(\frac{1}{10}\right) + 6C + 3\left(-\frac{1}{5}\right)$$

$$\rightarrow C = -2/5$$

Integrate

$$\int \left(\frac{3/10}{x-1} + \frac{1/10}{x+1} + \frac{(-2/5)x}{x^2+4} + \frac{(-1/5)}{x^2+4} \right) dx$$

$$= \frac{3}{10} \ln|x-1| + \frac{1}{10} \ln|x+1| - \frac{1}{5} \ln(x^2+4) - \frac{1}{10} \text{Arctan}\left(\frac{x}{2}\right) + K$$